



National University of Sciences and Technology (NUST)
School of Electrical Engineering and Computer Science

CS-212: Object Oriented Programming

Class: BESE 16B

Assignment 3

Due Date: April 8th, 2026

Instructor: Dr. Daud Abdullah

Total Marks: 30pts

Instructions

Code should be submitted in pdf format as 1 file for a group of 2 students groups.

Add comments along your code for justification.

Late submissions or submissions via email are **not** allowed.

Warning

Any copying directly **AI tools** of any sort would be considered plagiarism and would lead to zero marks being awarded for the assignment to both individuals.



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Objective: Create a SmartHome system where some components are built-in (Composition) and others can be moved in and out (Aggregation).

Identifying the Relationships

Before you code, look at these requirements and decide the relationship:

1. **SecuritySystem:** Every home comes with a built-in security panel. If the home is demolished, the panel is destroyed. (**Composition**)
2. **SmartAppliance:** Residents bring their own devices (like a Smart Coffee Maker or Vacuum). They take these with them when they move. (**Aggregation**)

The Implementation Task

1. The Component Classes

Create three simple classes with a powerOn() method:

- SecuritySystem: Prints "Security System Active."
- SmartAppliance: Has a name (e.g., "Toaster") and prints "[Name] is now running."

2. The SmartHome Class

This is where you demonstrate your architectural skills.

- **The Field:** Declare a SecuritySystem and a List<SmartAppliance>.
- **The Constructor:** Initialize the SecuritySystem **inside** the constructor.
- **The Method:** Create an add Appliance (SmartAppliance a) method to handle the aggregation.

Main Class

In your Main method, I want you to perform the following steps to prove the relationships:

1. **Create an Appliance:** Instantiate a SmartAppliance myCoffeeMaker = new SmartAppliance("Espresso Machine");.
2. **Build the Home:** Instantiate SmartHome myHome = new SmartHome();.
3. **Connect:** Add the coffee maker to the home.
4. **The "Demolition" Test:** * Print the coffee maker's status.
 - Set myHome = null; (Simulating demolition).
 - Try to print the coffee maker's status again.
 - *Answer the following:* Does the coffee maker still exist in memory? Can you still access the SecuritySystem?